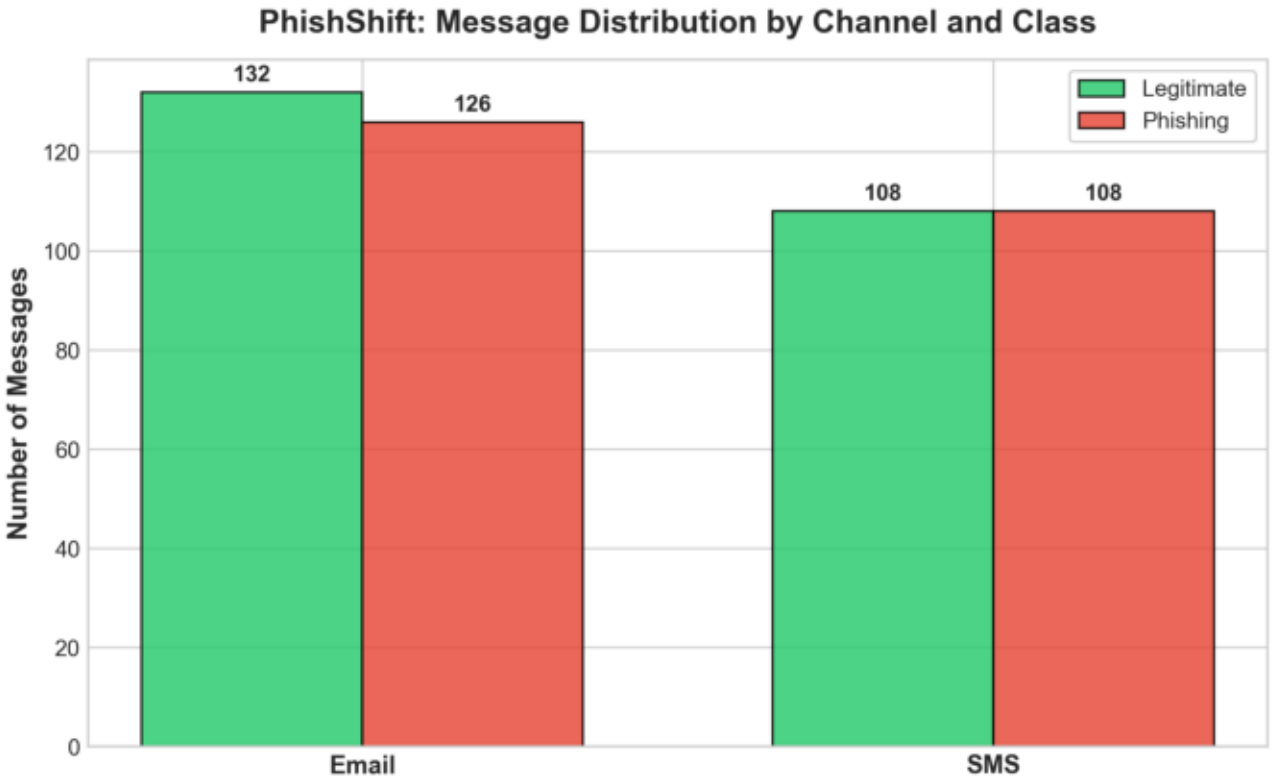
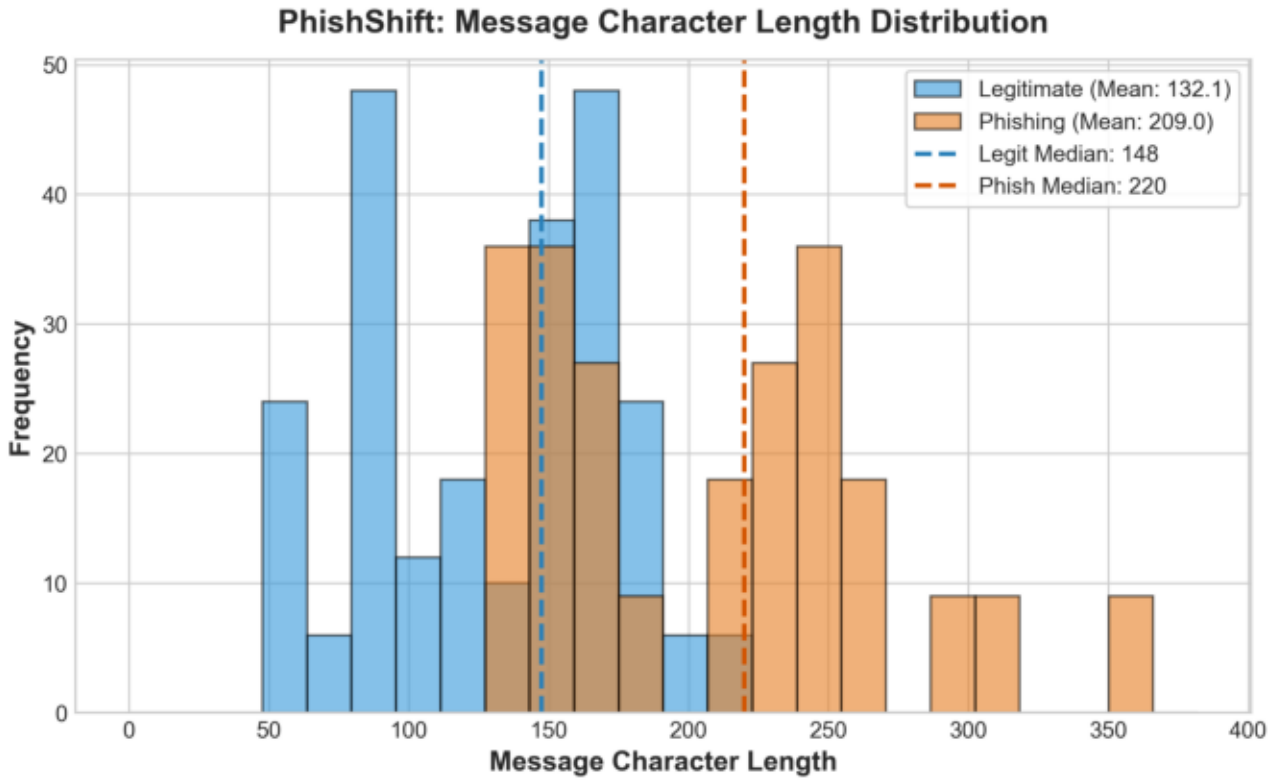
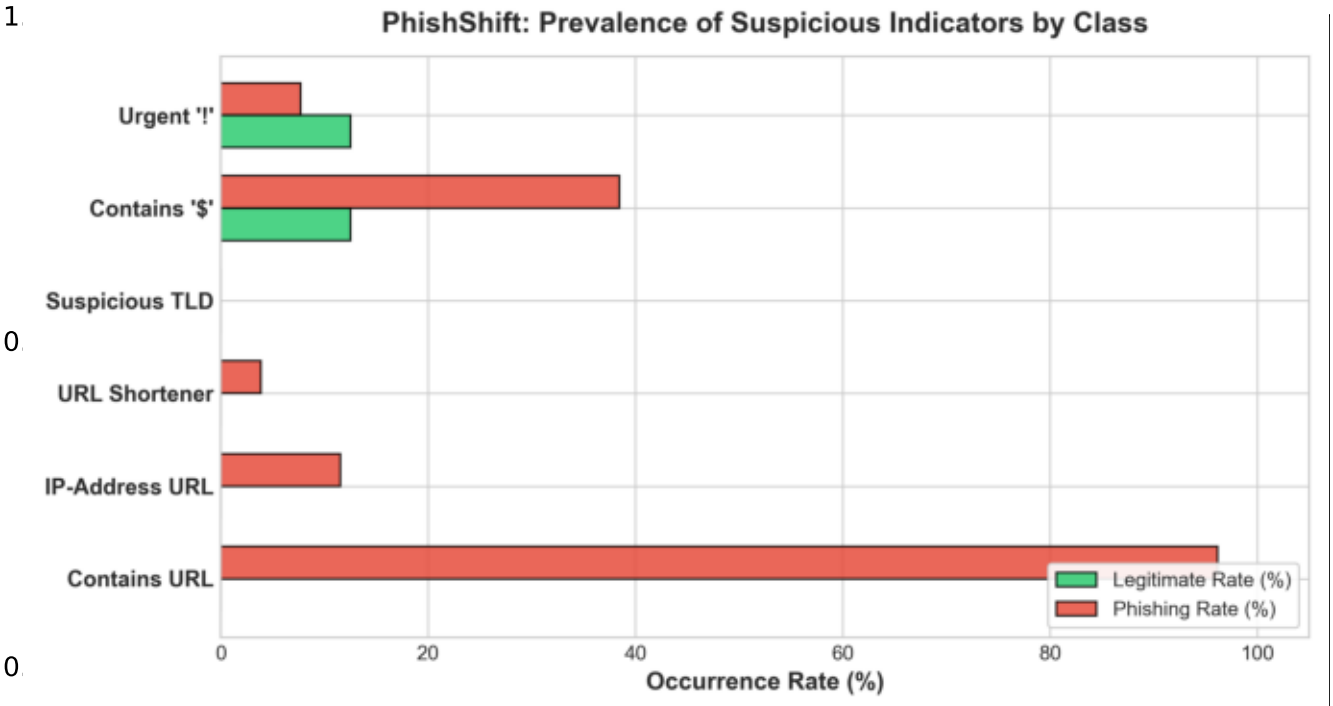
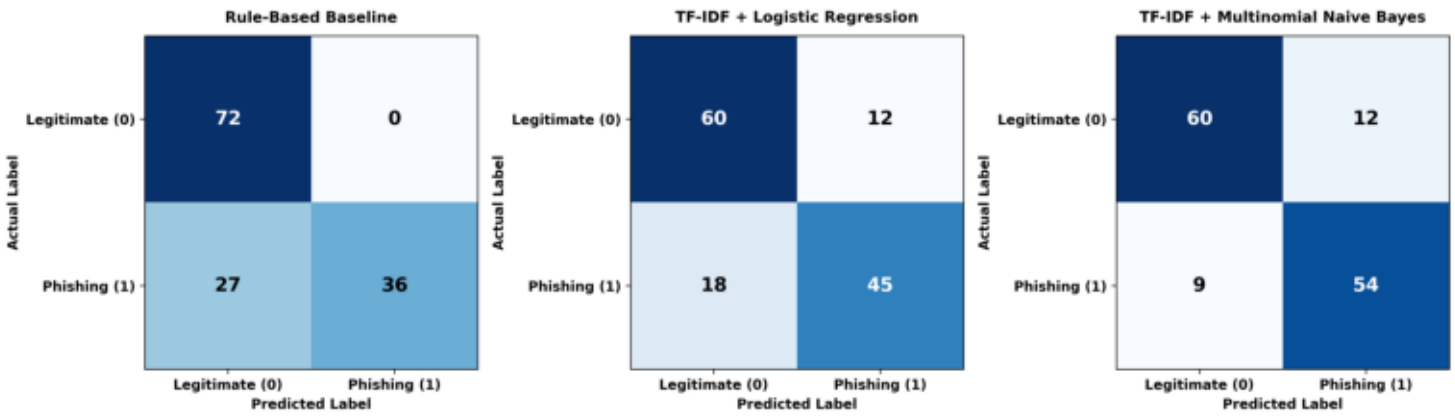
Cybersecurity Training Prototype & Machine Learning Technical Report


PhishShift: Message Distribution by Channel and Class
Number of Messages
Legitimate
Phishing
132
126
108
108
Email
SMS


PhishShift: Message Character Length Distribution
Frequency
Legitimate (Mean: 132.1)
Phishing (Mean: 209.0)
Legit Median: 148
Phish Median: 220
Message Character Length


PhishShift: Prevalence of Suspicious Indicators by Class
Urgent '!'
Contains '$'
Suspicious TLD
URL Shortener
IP-Address URL
Contains URL
Legitimate Rate (%)
Phishing Rate (%)
0
20
40
60
80
100
Occurrence Rate (%)


Rule-Based Baseline
Legitimate (0)
Phishing (1)
72
0
27
36
Actual Label
Legitimate (0)
Phishing (1)
Predicted Label
TF-IDF + Logistic Regression
60
12
18
45
TF-IDF + Multinomial Naive Bayes
60
12
9
54

0.0
0.2
0.4
0.6
0.8
1.0

5. Quantitative Evaluation & Boundary Case Analysis

MODEL BENCHMARK RESULTS (135 Unseen Held-Out Test Samples):

Model Architecture	Accuracy	Precision	Recall	F1-Score	FPR
Rule-Based Heuristic Baseline	0.8000	1.0000	0.5714	0.7273	0.0000
TF-IDF + Logistic Regression (Best)	0.7778	0.7895	0.7143	0.7500	0.1667
TF-IDF + Multinomial Naive Bayes	0.8444	0.8182	0.8571	0.8372	0.1667

DOCUMENTED BOUNDARY PREDICTION ERRORS:
1. CASE-01 (Operational Urgency False Positive):
   'Scheduled Server Maintenance Notice... Access will be temporarily restricted.'
   -> Model prob elevated by 'restricted'; mitigated to UNCERTAIN by operational context.

2. CASE-02 (Conversational BEC Gift Card Fraud):
   'Hi, are you at your desk? Quick favor. Purchase five $100 Apple gift cards... CEO.'
   -> Evaded bag-of-words text model; caught by heuristic executive/gift card scorer (70/100).

3. CASE-03 (Minimal-Context SMS 2FA):
   'Security code: 891240. Valid for 5 minutes.'
   -> Low token length resulted in borderline probability; safely abstained to UNCERTAIN.

ROBUSTNESS SUMMARY:
The model successfully withstood 8 adversarial transformations (all-caps, whitespace, punctuation, homoglyphs, synonym replacement, URL protocol casing). When character leetspeak (p@ssw0rd) was applied, the system safely shifted to UNCERTAIN rather than naively failing into a false negative.